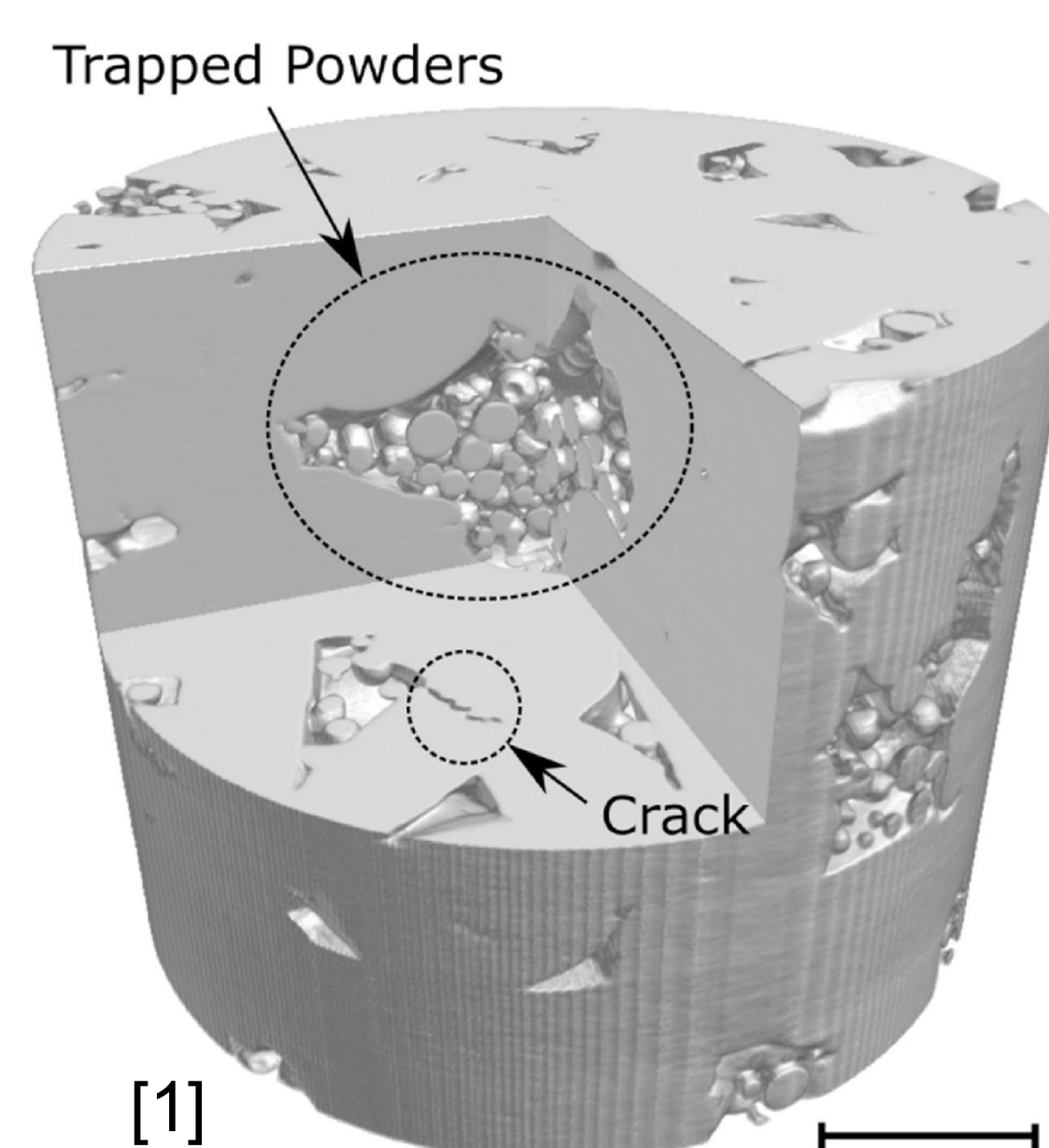




3D IMAGE SEGMENTATION FOR INTERNAL DEFECT IDENTIFICATION IN LPBF

BACKGROUND

- XCT is a pivotal tool for nondestructive evaluation of AM parts, but interpretation relies heavily on both practitioner knowledge and proprietary software.
- Images can provide insight such as defect distribution and morphology which are critical to mechanical performance.
- Many rely on broadly-applied local thresholds which have significant impact on the resulting segmentation.
- Little work has focused on the analysis of parameter-derived insights from defect data.



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ABSTRACT:

- This study presents an **automatic method** for the 3D segmentation and analysis of X-ray computed tomography (XCT) images and the application towards additively manufactured (AM) CoCr.
- Unsupervised machine learning** and **Voronoi-labeling** are proposed for 3D defect region segmentation.
- Density-based clustering** was used to group defects according to dimensions and mechanically significant morphology.
- The proposed methods utilize **open source tools** to develop 3D visualizations and analyze defect regions.

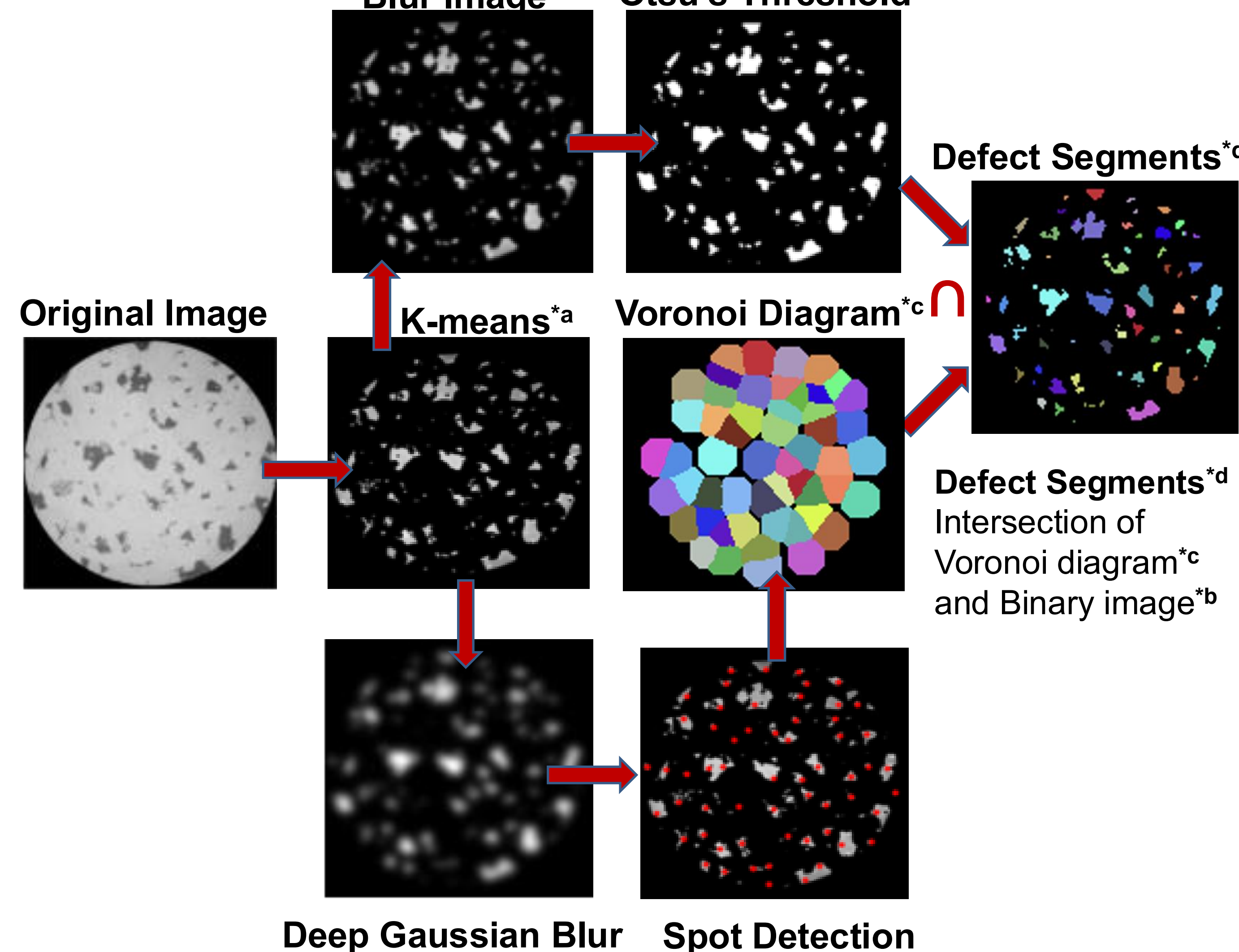
Voronoi Labeling^c

Subdivide defect regions starting from each expanding generating point that comes from spot detection step to form the Voronoi diagram expressed by:

$$R_k = \{x \in X | d(x, P_k) \leq d(x, P_j), \forall j \neq k\}$$

where each subregion in Voronoi diagram has a specific labeling index.

Slight Gaussian Blur Image
Binary Image by Otsu's Threshold^b



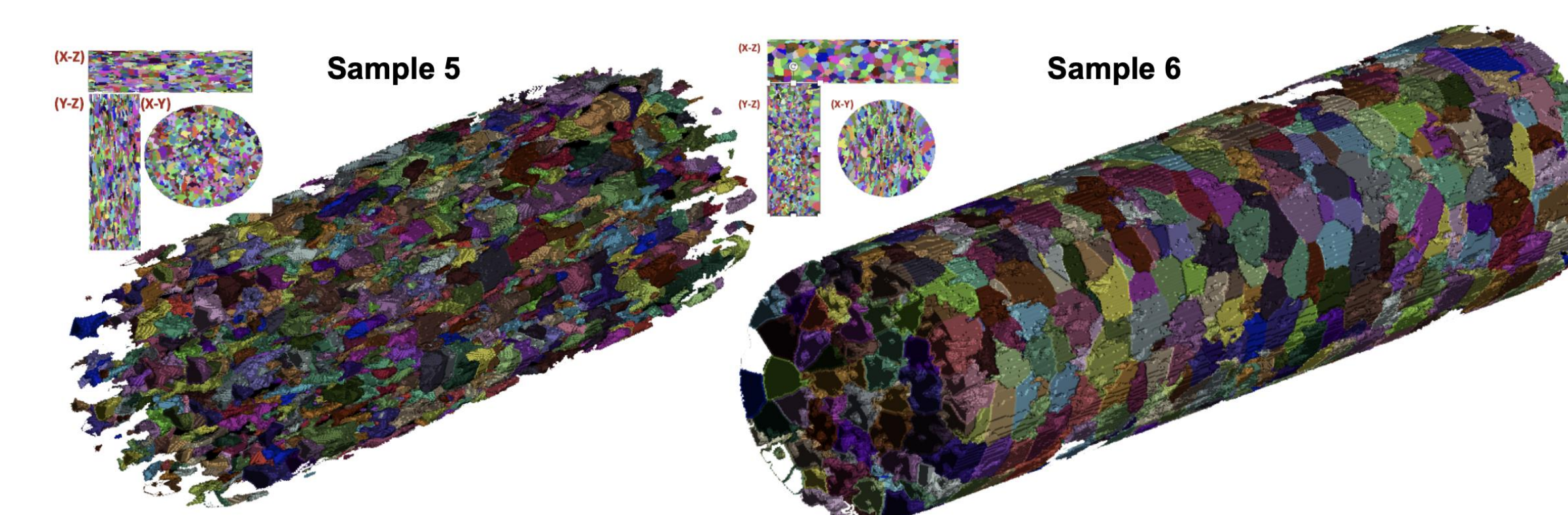
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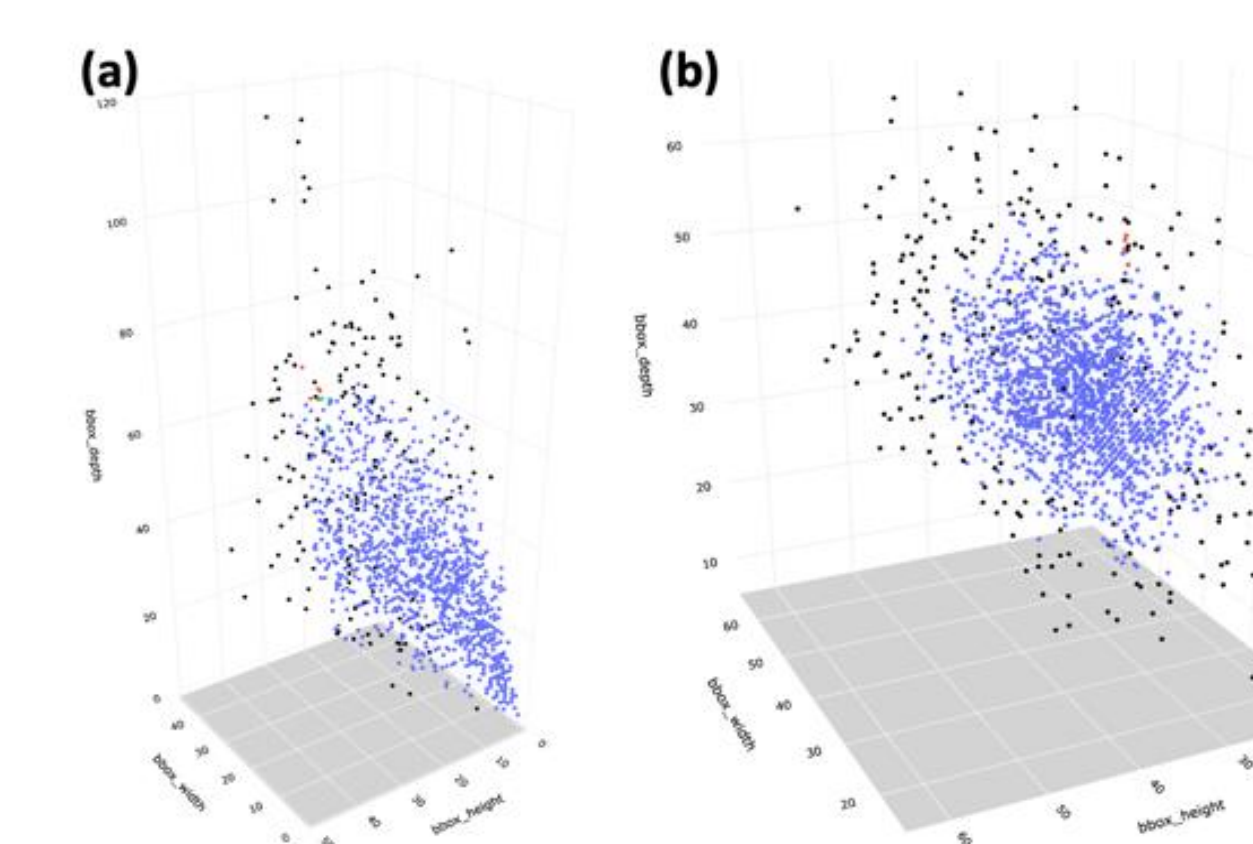
RESULT

1. 3D Defect Visualization

The defect regions for samples 5 and 6 are demonstrated in 3D space with the upper left corner corresponding to the segmented results for X-Y, X-Z, and Y-Z directions.



2. Density-based Spatial Clustering (DBSCAN)



Extract the feature vector $\vec{v} = [w, h, d]$ as the input for the DBSCAN algorithm, where parameters w, h, d are the width, height, and depth value of the bounding box of each 3D defect. (a) and (b) are the results of DBSCAN with parameters $\epsilon = 0.06$ and Mints=6 for samples 5 and 6, respectively.

DBSCAN identifies closely related defects as well as outliers. The blue points are typical defects while other colors represent outliers.

Clustering goodness may be measured by the silhouette coefficient.

3. Typical Defects Extraction

After identifying all outliers, we are left with defects that are most representative of the manufacturing process. By taking the average of the dimensional parameters $[w, h, d]$, the typical defect of each sample where its geometry is to the average value would be extracted and considered as the benchmark image for further analysis.

	(a)	(b)	(c)	(d)	(e)
Width:	4.3	23.5	8.1	15.3	35.6
Height:	4.7	23.2	10.6	14.9	35.6
Depth:	13.7	36.0	18.0	34.3	36.9

FUTURE WORK

- Integration of mechanically relevant morphological information (aspect ratio, sphericity) and manufacturing parameters into clustering algorithms for characterization and comparison.
- Quantifying defect criticality in terms of fatigue relevant properties such as "distance to surface" and morphology stress concentrators.
- Identify most relevant features for clustering to propose groups by potential cause (e.g. lack of fusion, gas trapped porosity).

ACKNOWLEDGEMENTS

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DATASET

- The CoCr AM XCT data is a publicly available dataset collected by NIST. The dataset consists of six sets of XCT images taken of laser powder bed fusion (LPBF) CoCr disks with no post-processing.
- Samples differed in processing parameters as shown in the table above.

METHODOLOGY

K-means Segmentation^a

Separate defect and solid regions by minimizing the sum of square distance between each cluster center and surrounding points.

$$J = \arg \min_s \sum_{i=1}^K \sum_{\mu_i \in S_i} \sqrt{|\mu_i - \tilde{\mu}_i|}$$

Otsu's Thresholding^b

Otsu's threshold minimizes the within-class variance of each class:

$$\hat{t} = \arg \min_t \{w_1(t)\sigma_1^2(t) + w_2(t)\sigma_2^2(t)\}$$

Pixel value at (x, y) of binary image:

$$f'_{x,y} = \begin{cases} 1, & f_{x,y} \leq \hat{t} \\ 0, & f_{x,y} > \hat{t} \end{cases}$$

